

Digital Maturity

GLOSSARY 01 - SHEET 3 / 8

The degree to which a firm has aligned technology, capability and governance to a digital operating model

PROFESSIONAL DEFINITION

Digital maturity is the extent to which an organisation has developed and aligned the technological assets, human capabilities, processes and governance mechanisms required to compete through digital means. It is expressed as a position on an ordered scale of stages rather than as a binary state, and it is measured across multiple dimensions simultaneously so that unevenness between them becomes visible (Kane et al., 2017; Chanias and Hess, 2016).

WHO DEFINED IT ORIGINALLY

The stage-model tradition originates with Nolan's (1973) stages of data-processing growth and was adapted for the digital era by Kane, Palmer, Phillips, Kiron and Buckley (2017) through the MIT Sloan Management Review and Deloitte global survey, and independently by Chanias and Hess (2016) at Ludwig-Maximilians-Universitat Munchen in a comparative review of maturity instruments.

WHY THIS DEFINITION WAS DEVELOPED

Boards asked how far along they were and received answers expressed in project counts. A maturity construct was needed to convert activity into position, to make comparison across business units and against competitors possible, and to expose the common pathology in which technology maturity runs ahead of talent and governance maturity.

HOW IT IS USED

Assess each dimension separately against stage descriptors, plot the profile, and direct investment at the lagging dimension rather than at the strongest one, because maturity advances only at the rate of its weakest component.

WHEN IT IS USED

As a baseline before a transformation programme, annually thereafter to track movement, and at post-merger integration to compare the maturity of the two organisations being combined.

BY WHOM IT IS USED

Transformation programme offices running the assessment, executive committees allocating investment between dimensions, and business unit leaders benchmarking against the group profile.

FIGURE 1 - THE MATURITY STAIRCASE

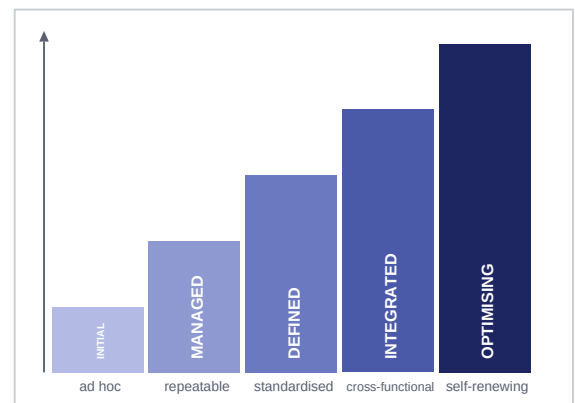


Figure 1. Maturity as an ordered progression. Each stage presupposes the one below it, and a firm advances only at the rate of its least mature dimension - the reason technology-first programmes stall at the third step (after Kane et al., 2017).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Nolan's original model assumed a single sequence through which every organisation passed at its own speed, and was criticised for weak empirical support. The digital-era models abandoned the single sequence for a multi-dimensional profile: a firm can be advanced in technology and immature in talent, and it is precisely that asymmetry that predicts failure. Maturity thereby became a diagnostic instrument rather than a scorecard.

RELATED DEFINITIONS IN THIS COURSE

[Digital Transformation - Unit 1](#)
[Technology Adoption Lifecycle - Unit 1](#)
[AI Adoption - Unit 4](#)
[AI Roadmap - Unit 4](#)
[Organisational Culture - Unit 2](#)

SOURCES (HARVARD REFERENCING STYLE)

- Chanias, S. and Hess, T. (2016) 'How digital are we? Maturity models for the assessment of a company's status in the digital transformation', Management Report 2/2016. Munich: LMU Institute for Information Systems and New Media.
- Kane, G.C., Palmer, D., Phillips, A.N., Kiron, D. and Buckley, N. (2017) Achieving Digital Maturity. Cambridge, MA: MIT Sloan Management Review and Deloitte University Press.
- Nolan, R.L. (1973) 'Managing the computer resource: a stage hypothesis', Communications of the ACM, 16(7), pp. 399-405.
- Teichert, R. (2019) 'Digital transformation maturity: a systematic review of literature', Acta Universitatis Agriculturae et Silviculturae Mendelianae Brunensis, 67(6), pp. 1673-1687.